

Saksham Arora

Masters in Computer Science & Statistics

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Education

Purdue University

Masters in Statistics & Computer Science

West Lafayette, IN, USA

Aug 2026

Indian Institute of Technology Kanpur

Major: Chemical Engineering, Minors: Machine Learning & Computer Systems

Kanpur, India

Nov. 2020 - May 2024

GPA: 8.7/10.0

Coursework

[Click on the course to view details](#)

Computer Science	Fundamentals of Computing, Data Structures and Algorithms, Introduction to Machine Learning, Topics in Learning Theory, Image Processing, Computer Networks, Blockchain and its Applications, Computational Methods in Engineering, Algorithm Toolbox
Mathematics	Linear Algebra, Ordinary Differential Equations, Real Analysis, Probability & Statistics
Electives	Introduction to Electronics, Data Science Practical

Research Experience

Bilevel Reinforcement Learning for Circularity in Ammonia-Urea Manufacturing

Prof. Vaneet Aggarwal, Edwardson School of Industrial Engineering, Purdue University, USA

May 2026 – Present

- Formulated waste-pricing coordination in a 3-factory ammonia-urea production network as a bilevel reinforcement learning problem, with a price-setting leader and best-responding follower agents, addressing a coordination failure inherent to symmetric multi-agent RL
- Built a reproducible simulation environment from first-principles process models (electrolysis, Haber–Bosch, urea synthesis) with exact mass-balance closure; generated 414K samples via orthogonal Latin hypercube sampling and trained neural surrogates achieving $R^2 > 0.998$ in- and out-of-distribution
- Implemented and adapted a penalty-based bilevel hypergradient algorithm (PP0 with GAE, six modifications for multi-stage, high-dimensional continuous control), improving circular-economy waste recovery by 8.5 percentage points over symmetric multi-agent RL baselines ($p < 0.01$, 10-seed evaluation)
- Diagnosed a market-calibration failure mode where the leader's pricing action had no influence on follower behavior, and a reward-design pitfall where an explicit circularity bonus increased the optimization objective while reducing physical recovery
- Conducted statistical validation (paired t -test, Wilcoxon signed-rank, Bonferroni correction, pre-registered hypotheses) across 10 random seeds; manuscript in preparation for *IEEE Transactions on Industrial Informatics*

Numerical Modelling of Debris Flow Movement | Mitacs Globalink Research Internship

Prof. Dwayne Tennant, School of Engineering, The University of British Columbia, Canada

May. 2023 – Jul. 2023

- Developed a machine learning pipeline that predicts volume of debris flow triggered by rains post wild-fires
- Built and optimized Support Vector Machine (SVM) models, improving reliability over baseline statistical methods
- Automated hydrograph and rainfall plot generation via web scraping using Flask for efficient data-driven visualizations
- Utilized Flask and React to deploy the model, enabling public access for constructing life-saving barriers

Gen4Edu | AI-Enhanced Knowledge Graphs for NCERT-Based Learning | Report

NLP Lab, Department of Computer Science and Engineering, IIT Kanpur

Jan. 2024 – Apr. 2024

- Develop a study aid that generates outcomes akin to ChatGPT, but tailored to a knowledge base from curriculum books
- Engineered an automated pipeline to extract entities from NCERT texts using PyPDF and 4-bit quantized Mistral-7B
- Designed zero-shot prompts to extract ontology triplets, utilizing NetworkX for graph structuring and community detection
- Implemented a semantic evaluation framework using all-MiniLM-L6-v2 and Cosine Similarity to benchmark response relevance

Beyond Borders: A data-driven study of postgraduate student mobility

Prof. Dootika Vats, Department of Mathematics and Statistics, IIT Kanpur

Aug. 2023 – Nov. 2023

- Analyzed postgraduate applications using RStudio, uncovering trends in foreign university admissions
- Implemented robust data scraping pipelines with rvest and RSelenium, ensuring precise application data collection
- Optimized preprocessing workflows using dplyr, incorporating techniques to mitigate potential sample bias
- Developed interactive visualizations using ggplot2, plotly, and Shiny, providing insights into major preferences, admit probabilities, and academic trajectories

Particle Simulator using CUDA

Prof. Anurag Tripathi, Department of Chemical Engineering, IIT Kanpur

Jan. 2023 – Apr. 2023

- Implemented a massively parallel particle simulator capable of simulating 1 Million+ particles in real time
- Optimized collision detection from $O(N^2)$ to $O(N)$ using uniform spatial grid (Spatial Hashing) to localize the interactions
- Maximized memory throughput by utilizing Thrust library to sort particles by Grid Index, ensuring coalesced memory access

Attack Detection on a COTS Drone

Prof. Urbi Chatterjee, Dept. of CSE, IIT Kanpur

May 2022 – Jul. 2022

- Simulated adversarial drone attacks using aircrack-ng, hping3, and Wireshark on Kali Linux generating labeled traffic data
- Captured and parsed .pcap traces; extracted features such as packet inter-arrival times, RSSI variance using Scapy
- Trained and evaluated an SVM classifier to distinguish benign drone communication from manipulated control signals, completing an end-to-end ML workflow from data generation to model development

Work Experience

Flipkart (Indian Walmart Subsidiary), Bengaluru, India | Business Analyst

Principal Homepage Analyst

Jul. 2024 – Present

- Engineered a custom ReAct (Reasoning + Acting) agentic application powered by Gemini 2.5 Pro, enabling the LLM to autonomously sequence Python-based analytical tools, evaluate intermediate KPI fluctuations, and iteratively deduce root causes analysis without human intervention.
- Developing controlled A/B experimentation frameworks for homepage personalization using uplift modeling, identified domains where randomized trials fail and engineered Causal Inference pipelines using Double Machine Learning (DML) to ensure unbiased treatment-effect estimation
- Led the launch of the new homepage experience for low-propensity users, driving a 3% increase in Page Views, and optimized journey-continuation widgets, resulting in a 5% uplift in homepage revenue
- Developed robust time-series forecasting pipelines using Facebook Prophet, ARIMA, and STL decomposition with automated hyperparameter tuning and cross-validation to reliably predict traffic and engagement trends
- Prototyped a real-time feed refresh model using XGBoost with feature engineering on recency, affinity, and scroll-depth signals; optimized latency for high-throughput production environments
- Designed and introduced novel engagement and ranking metrics (*Homepage Redundancy*, *NDCG-based scoring*) enabling richer diagnostic capabilities and improved evaluation of ranking performance beyond traditional CTR-based metrics
- Automated real-time monitoring dashboards using Python (Pandas, Plotly) and BI tools; integrated Kafka-based streaming data pipelines for hourly tracking of traffic, experimentation, and sales metrics during *Big Billion Days*

Achievements and Accolades

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|------|---|
| 2025 | Awarded with <i>Kool and the Gang</i> award at Flipkart on demonstrating great collaboration and operational excellence |
| 2024 | Received tuition-fee grant for four consecutive years for excellent performance in overall academics and extra-curricular |
| 2023 | Secured Gold Medal in Street Photography at Inter IIT Cultural Meeting |
| 2021 | Awarded Academic Excellence Award in the year 2020-2021 |
| 2020 | National top 1% in JEE Advanced 2020 among the 2.3 lakhs shortlisted candidates from JEE Mains |
| 2019 | Received scholarship from MHRD, Government of India for being in top 5 percentile in Grade XII (High School) in India |

Leadership and Volunteering Experience

Academic Mentor: Mentored incoming freshmen on core institute courses for Linear Algebra and ODE & Intro to Programming

Club Head - Photography Club: Led a 30+ member team, promoting photography in the campus community through social media, exhibitions, and competitions, and managing an INR 1,00,000 budget for equipment and events

Secretary – Electronics Club: Responsible for conducting lectures, workshops, and maintaining records of club activities. Also, designed social media content and managed club handles.

Prayas – IIT Kanpur: Volunteered by providing classes on basic maths to children from villages around IIT Kanpur

Technical Skills

Languages and Tools: Python, C/C++, R, SQL, MATLAB, Bash, Solidity, Git, Selenium, BeautifulSoup

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, XGBoost, OpenCV, Pandas, NumPy, Prophet, DML

Systems: Django, Flask, React.js, Express.js, PostgreSQL, CUDA, Hyperledger Fabric

Technical Projects

Payroll System using Hyperledger Fabric

Course Project, Prof. Angshuman Karmakar, IIT Kanpur

Jan. 2024 – Apr. 2024

- Developed blockchain-based infrastructure for cross-border payments using Ethereum and Solidity smart contracts
- Implemented progressive payroll system with Django-Python, featuring employee and employer dashboards
- Utilized Hyperledger Fabric for enterprise-grade blockchain deployment, ensuring data privacy & permissioned access
- Integrated banking APIs for seamless financial transactions using RESTful architecture

3D Pose Warping

Electronics Club, IIT Kanpur

May. 2021 – Jul. 2021

- Developed a deep learning pipeline to manipulate human poses in images and align them with reference images
- Implemented a Generative Adversarial Network for the Fashion3D dataset using the ResNet-50 architecture
- Achieved SSIM values of 0.9346 (background) and 0.3464 (foreground) after 100 epochs of training

Audio Classification using Neural Networks

Robotics Club, IIT Kanpur

Nov. 2022 – Jan. 2023

- Leveraged spectrogram features for Automatic Speech Recognition and trained neural networks using TensorFlow
- Employed diverse preprocessing techniques in Python to enhance feature extraction for model training
- Achieved 87.8% accuracy using CNN, outperforming a DNN baseline accuracy of 74.9%